

Required Software Tools

Vehicle Parking System — Group 14 | Setup guide for Windows 11

1. Version Control & Collaboration

Git for Windows — includes Git Bash and the git CLI — git-scm.com/download/win

GitHub account — one per team member, added as a collaborator on the shared repository — github.com

GitHub Desktop — optional GUI for Git, useful if you'd rather avoid the command line — desktop.github.com

2. Node.js Runtime

Node.js 24 (Active LTS) — JavaScript runtime; the installer includes npm automatically — nodejs.org/en/download

nvm-windows — Node Version Manager for Windows — installs and pins the exact same Node.js version on every machine so builds behave identically — github.com/coreybutler/nvm-windows/releases

Recommended: install nvm-windows first, then use it to install Node.js 24 (`nvm install 24` then `nvm use 24`), instead of installing Node directly.

3. Database

MongoDB Community Server — latest stable release, 8.x series — mongodb.com/try/download/community

MongoDB Compass — official GUI for browsing and querying the database; offered as an install option in the Community Server installer — mongodb.com/products/compass

4. Code Editor

Visual Studio Code — recommended editor for the project — code.visualstudio.com

5. API Testing

Postman — for manual/exploratory testing of API endpoints during development — postman.com/downloads

6. Automated Testing

Jest — test runner used for backend and frontend unit/integration tests

Installed automatically via **npm install** once the project repository is cloned — not a separate download.

Supertest — HTTP-assertion library for testing Express API endpoints, used inside Jest

Installed automatically via **npm install** — not a separate download.

React Testing Library — component-level testing for the React frontend, used inside Jest

Installed automatically via **npm install** — not a separate download.

7. Load / Concurrency Testing

k6 — Grafana's load-testing tool, used for the multi-gate concurrency test — k6.io

Install via Windows Package Manager: open PowerShell and run `winget install k6 --source winget`